

Mohammadreza Hajibabaei

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Education

Sharif University of Technology, M.Sc. in Electrical Engineering – Digital Systems

Sept 2024 – present

- Admitted without national entrance exam among the top 30% of Sharif B.Sc. students, based on academic excellence.
- **Thesis(defined Sept 2025):** Efficient Allocation of Edge Computing Resources for Pipelined Execution of Deep Learning Models
- Advisor: Prof. Hamed Shah Mansouri
- Keywords: Edge Computing, Deep Learning, Resource Allocation, Multi-Objective Optimization
- **Coursework:** Deep Learning, Deep Generative Models, Neural Networks, Convex Optimization, Massive Data Analytics

Sharif University of Technology, BS in Electrical Engineering

Sept 2020 – May 2024

- **Coursework:** Introduction to Machine Learning, ASIC/FPGA System Design, FPGA Systems Lab, Machine Learning and Vision Lab, AI & Biological Computation, Data Structures and Algorithms, Parallel Programming & architecture

Research Interests

- My work focuses on improving the efficiency of deep learning inference in distributed environments through intelligent partitioning of DNN models across TPU and edge devices. Interests include resource optimization, hardware–software co-design, and scalable edge computing architectures for AI applications.

Experience

M.Sc. Thesis – Sharif University of Technology

Sept 2025 – Present

- Title: Efficient Allocation of Edge Computing Resources for Pipelined Execution of Deep Learning Models
- Advisor: Prof. Hamed Shah-Mansouri
- Awarded competitive research grant for TPU-based edge AI system design (Sharif University Research & Technology Vice-Presidency).
- Researching intelligent DNN partitioning across TPUs and edge devices to optimize latency and resource efficiency.

Research Intern, Kavoshcom Company

June 2023 – Aug 2023

- Developed and evaluated machine learning algorithms for classification of ECG signals and detection of cardiac arrhythmias.
- Applied signal preprocessing, feature extraction, and supervised learning techniques to improve diagnostic accuracy.
- Tools: Python, scikit-learn, NumPy, Keras

Undergraduate Research Assistant, ACL Lab (Prof. Bagheri), Sharif University of Technology

Sept 2023 – Aug 2024

- Collaborated with graduate researchers on machine learning–based audio analysis projects.
- Gained exposure to research related to context-aware audio annotation and classification, including model training, feature extraction, and evaluation techniques

Teaching Assistant, Mathematical Methods in Engineering ,Sharif University of Technology

Sept 2024 – Jan 2025

- Assisted Prof. Reza Kazemi and Prof. Hossein Pilaram in teaching, holding tutorials, and grading assignments.
- Supported students in understanding applied linear algebra and matrix computations.

Projects

AI Chat Bot — Sharif University of Technology — Deep Learning Course, Spring 2025

- Developed an intelligent conversational AI agent with real-time reasoning, web-integrated responses, and memory-based dialogue management using Retrieval-Augmented Generation (RAG) and function calling.
- Implemented document understanding, 20-Questions interaction, and GUI testing via Gradio, demonstrating skills in LLM integration, prompt engineering, and deep learning systems (Python, PyTorch, Transformers, FAISS, PyMuPDF).

FPGA-Based Signal Processing and Communication System (Zynq Platform), Spring 2024

- Designed and implemented a real-time signal processing system on a Zynq SoC (PS+PL). Implemented waveform generation, FFT visualization on HDMI, and a scrambler/descrambler link following DVB-S2 standard.
- Demonstrated advanced FPGA design, signal processing, and embedded system integration using AXI DMA, FFT IP, and HDMI controllers.

Skills

Programming: Python, C/C++, Verilog, Java

Simulation / Data Tools: MATLAB, OMNeT++, PySpark, Git, Cuda

Hardware Tools: Vivado, Xilinx ISE, Quartus, ModelSim

Frameworks: Numpy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenAI, Transformers